

Explain Bit, Tit, Btit, Rbtit, Rtit, Fuzzy Number system

Bit: Binary representation using 0 and 1.

Tit: Ternary representation using 0, 1, and 2.

Btit: Balanced Ternary

Rbtit: Redundant Balanced Ternary

Rtit: Redundant Ternary

Fuzzy: Based on degrees of certainty.

$$1-(111)_{(2)}=?_{(10)} \quad 1 * 2^0 + 1 * 2^1 + 1 * 2^2 = 7$$

$$2-(101 \ 111)_{(2)}=?_{(8)} = 57$$

$$3-(1001 \ 0011 \ 0110)_{(2)}=?_{(16)} = 936$$

$$4-(1010 \ 1011 \ 0111)_{(2)}=?_{(16)} = AB7$$

$$5-(101)_{(2)}=?_{(10)} = 1 * 2^2 + 0 * 2^1 + 1 * 2^0 = 5_{(10)}$$

$$6-(1 \ 101)_{(2)}=?_{(8)} = 15_{(8)}$$

$$7-(111 \ 101 \ 110 \ 011)_{(2)} = 7563_{(8)} = F73_{(16)}$$

$$8-(1111 \ 0111 \ 0011)_{(2)}=?_{(16)} = F73_{(16)}$$

$$9-574_{(8)} = ?_{(2)} = 110 \ 111 \ 100_{(2)}$$

$$10-ABC_{(16)} = ?_{(2)} = 1010 \ 1011 \ 1100_{(2)}$$

$$11-ABC_{(16)} = 5274_{(8)} =$$

$$12- \quad 12_{(2)} = 4_{(10)} = 1 * 2^1 + 1 * 2^0 = 4_{(10)}$$

$$13- \quad 20_{(2)} = 4_{(10)} = 2 * 2^1 + 0 * 2^0 = 4_{(10)}$$